

Viral obesity: fact or fiction?

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The aetiology of obesity is multifactorial. An understanding of the contributions of various causal factors is essential for the proper management of obesity. Although it is primarily thought of as a condition brought on by lifestyle choices, recent evidence shows there is a link between obesity and viral infections. Numerous animal models have documented an increased body weight and a number of physiologic changes, including increased insulin sensitivity, increased glucose uptake and decreased leptin secretion that contribute to an increase in body fat in adenovirus-36 infection. Other viral agents associated with increasing obesity in animals included canine distemper virus, rous-associated virus 7, scrapie, Borna disease virus, SMAM-1 and other adenoviruses. This review attempted to determine if viral infection is a possible cause of obesity. Also, this paper discussed mechanisms by which viruses might produce obesity. Based on the evidence presented in this paper, it can be concluded that a link between obesity and viral infections cannot be ruled out. Further epidemiologic studies are needed to establish a causal link between the two, and determine if these results can be used in future management and prevention of obesity.